

Problem Set Solutions 13

Comp 301

Spring 2025

Week 13: 26.05.2025 - 30.05.2025

Problem 1: I

- (1) For the programs below draw the environment and store diagram both for Call by Need and Call by Name before and after the **red** expressions are evaluated.

(a) let **a=23** in let f=proc(x) begin x; x; x end in (f begin set a=-(a, -10); a end)

Call by Need:

Before

Environment:

((i 0) (v 1) (x 2))

Store:

0: (struct: num-val 1) 1: (struct: num-val 5) 2: (struct: num-val 10)

After

Environment:

((i 0) (v 1) (x 2) (a 3))

Store:

0: (struct: num-val 1) 1: (struct: num-val 5) 2: (struct: num-val 10)

3: (struct: num-val 12)

Call by Name

Before

Environment:

((i 0) (v 1) (x 2))

Store:

0: (struct: num-val 1) 1: (struct: num-val 5) 2: (struct: num-val 10)

After

Environment:

((i 0) (v 1) (x 2) (a 3))

Store:

0: (struct: num-val 1) 1: (struct: num-val 5) 2: (struct: num-val 10)

3: (struct: num-val 12)

(b) let a=23 in let f=proc(x) begin x; x; x end in (f begin set a=-(a, -10); **a** end)

Call by Need

Before

Environment:

((f 4) (a 3) (i 0) (v 1) (x 2))

Store:

0: (struct: num-val 1) 1: (struct: num-val 5)

2: (struct: num-val 10)

3: (struct: num-val 25)

4: procedure x ... ((a 3) (i 0) (v 1) (x 2))) 5: (thunk ... ((f 4) (a 3) (i 0) (v 1) (x 2)))

After

Environment:

((x 5) (a 3) (i 0) (v 1) (x 2))

Store:

0: (struct: num-val 1) 1: (struct: num-val 5)

2: (struct: num-val 10)

3: (struct: num-val 25)
 4: procedure x ... ((a 3) (i 0) (v 1) (x 2)))

Call by Name

Before

Environment:
 ((f 4) (a 3) (i 0) (v 1) (x 2))

Store:
 0: (struct: num-val 1) 1: (struct: num-val 5)
 2: (struct: num-val 10)
 3: (struct: num-val 38)
 4: (procedure x ... ((a 3) (i 0) (v 1) (x 2))) 5: (thunk ... ((f 4) (a 3) (i 0) (v 1) (x 2)))

After

Environment:
 ((x 5) (a 3) (i 0) (v 1) (x 2))

Store:
 0: (struct: num-val 1) 1: (struct: num-val 5)
 2: (struct: num-val 10)
 3: (struct: num-val 51)
 4: (procedure x ... ((a 3) (i 0) (v 1) (x 2))) 5: (thunk ... ((f 4) (a 3) (i 0) (v 1) (x 2)))

(2) **letrec** infinite-loop(c)=(infinite-loop -(c, 1)) in **let** f= **proc**
 (x) **begin** -(2, 1); 10 **end** in (f (infinite-loop 99))

(a) What would happen if you ran the program above in IREF? **Program gets stuck in an infinite loop**

(b) Did you observe a difference between Call by Name and Call by Need when evaluating the above program, explain the reason briefly. **No.**

(c) For the program above draw the environment and store diagrams during **(f (infinite-loop 99))** is evaluated.